SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA0202

COURSE OUTCOME COVERED
CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 3 Total Marks:30 Annexure A

Scenario-Based

Code of Conduct:

I M Tilak Venkata Sai(192424415)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

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Criteria	**Weight age (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Understa nding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identifi ca tion of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Applic ati on of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise.

Questions:

A. Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

B. Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

C. After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

Questions:

A. Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

B. Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

C. If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

B. Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

1. Crack Detection on Concrete Surfaces Using Drone Images

A. Image preprocessing pipeline

1. **Image Acquisition** – Capture high-resolution drone images.
2. **Illumination Correction** – Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) or Histogram Equalization to reduce uneven lighting and shadows.
3. **Noise Reduction** – Use a Gaussian filter or Bilateral filter to remove Gaussian noise while preserving edges.
4. **Contrast Enhancement** – Enhance the visibility of thin cracks using CLAHE or contrast stretching.
5. **Image Normalization** – Normalize pixel intensity values to reduce brightness variations between images.
6. **Grayscale Conversion** – Convert RGB images to grayscale for easier crack analysis.

Justification:

- Illumination correction compensates for changing sunlight and shadows.
- Noise reduction removes Gaussian noise caused by camera vibration.
- Contrast enhancement makes thin cracks more visible.
- Normalization provides consistent intensity values for reliable processing.
- Grayscale reduces computational complexity.

B. Suitable edge detection technique

Canny Edge Detection is the most suitable.

Comparison with Sobel:

Canny	Sobel
Better noise suppression	More sensitive to noise
Detects thin and weak edges	Misses fine cracks
Produces continuous edges	Produces broken edges
Uses non-maximum suppression	No edge thinning

Reason: Canny detects fine cracks accurately while minimizing false edges.

C. Post-processing technique

Use **Morphological Closing (Dilation followed by Erosion)**.

Benefits:

- Connects broken crack segments.
- Fills small gaps.
- Produces continuous crack boundaries.
- Improves crack detection accuracy.

2. Human Motion Tracking in a Security Camera Video

A. Lucas-Kanade Optical Flow

Lucas-Kanade estimates the motion of feature points between consecutive video frames.

Working:

1. Detect feature points (corners).
2. Assume small movement between frames.
3. Compute displacement using neighboring pixels.
4. Generate optical flow vectors showing motion direction and speed.

It works efficiently for real-time human tracking.

B. Aperture Effect

The aperture effect occurs when motion is observed through a small local region.

Effects:

- Uniform walls and floors have very few texture features.
- Clothing with smooth surfaces also lacks distinctive patterns.
- Motion direction cannot be estimated accurately.
- Optical flow vectors become unreliable or ambiguous.

C. Sudden change in speed or direction

Effects:

- Optical flow vectors change suddenly in magnitude and direction.
- Tracking errors may occur.
- Person may temporarily be lost.

Improvement Method:
Use the **Kalman Filter** (or Particle Filter) to predict the person's next position and maintain stable tracking during sudden movements.

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3. Vehicle Tracking at a Traffic Junction Using Video Frames

A. Optical flow for vehicle tracking

Optical flow calculates the movement of vehicles between consecutive frames.

Steps:

1. Detect moving vehicle features.
2. Compute optical flow vectors.
3. Determine motion direction from vector orientation.

4. Estimate vehicle speed from vector magnitude.
5. Track vehicles across multiple frames.

B. Two challenges affecting motion estimation

1. Occlusion

- Vehicles overlap and hide each other.
- Some motion vectors disappear.
- Tracking may switch from one vehicle to another.

2. Shadows

- Shadows are detected as moving objects.
- Creates false optical flow vectors.
- Reduces tracking accuracy.

(Other acceptable challenges include sudden braking or camera vibration.)

C. Improvement method

Use **Kalman Filter with Object Detection (e.g., YOLO + Kalman Filter)**.

Advantages:

- Predicts vehicle position during occlusion.
- Handles sudden direction changes.
- Maintains continuous tracking.
- Reduces tracking errors caused by overlapping vehicles.
- Improves overall tracking reliability.